CO2xNxPhrag: Porewater Ammonium

July 2026 Samples

2026-10-19

Contents

0.1 Assessing standard Curves . . . 4
0.2 Update Standard Log . . . 4
0.3 Dilution Corrections - ensure the latest dilution is kept . . . 6
0.4 Performance Check . . . 6
0.5 Analyze the Check Standards . . . 7
0.6 Analyze Blanks . . . 8
0.7 Analyze Duplicates . . . 9
0.8 Spikes . . . 10
0.9 Matrix Effects . . . 11
0.10 Create Log for QAQC . . . 12
0.11 Merge samples with Metadata & check all samples are present . . . 15
0.12 Organize Data . . . 16
0.13 Visualize Data . . . 16

##Run Information

```
cat("Run Information: ") #lets you know what section you're in
```

```
## Run Information:
```

```
library(here)
```

```
## here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW
```

```
#set the run date & user name
  sample_year <- "2026"
  sample_month <- "July"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/20260908_Phrag_July2026.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- here("Porewater QAQC Logs", "GCReW_NH4_QAQC_Log.csv")
  file_path_QAQC <- here("Porewater QAQC Logs", "GCReW_SEAL_Check_QAQC_Log.csv")
  final_path <- "Processed Data/Phrag_Porewater_NH4_202607.csv"

#record any notes about the run or anything other info here (Leave empty otherwise):
  run_notes <- ""


  cat(run_notes)
```

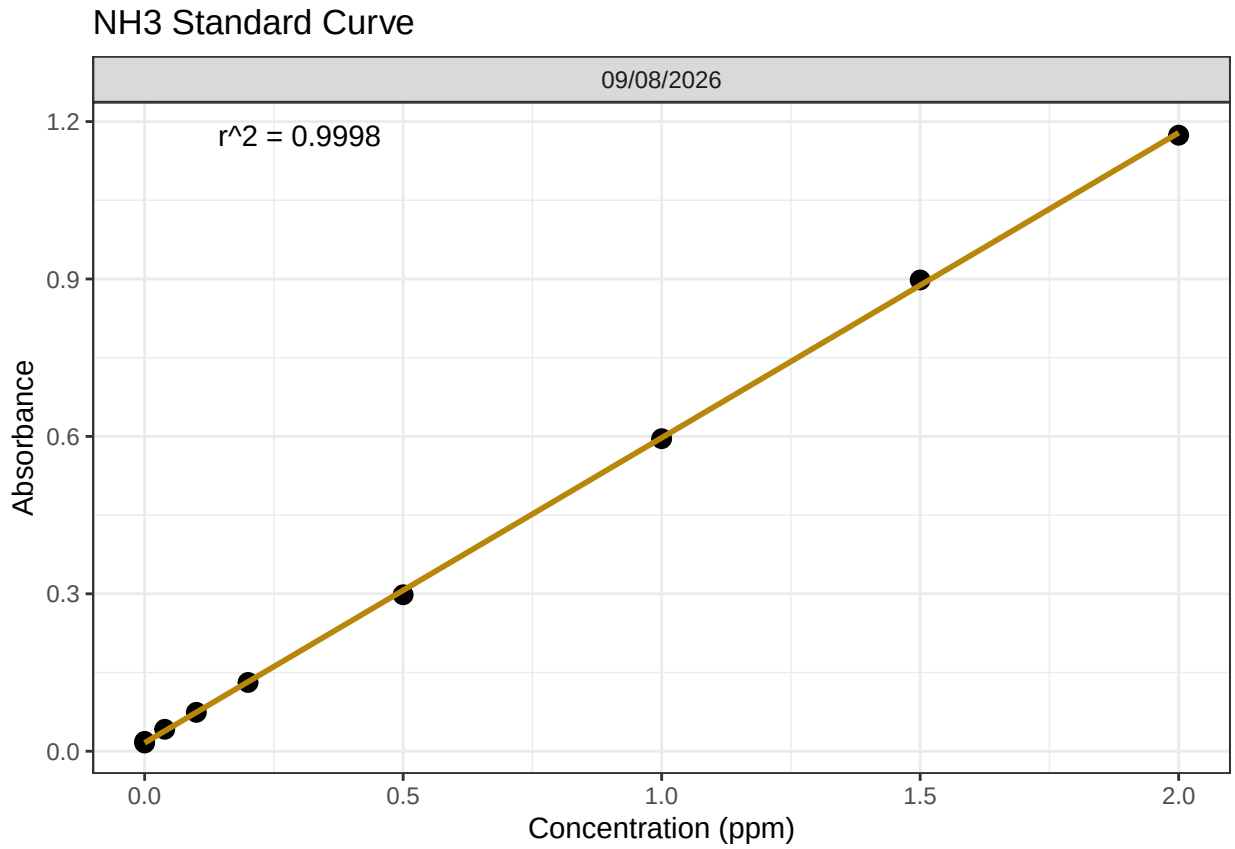
##Setup

##Read in metadata and create similar sample IDs for matching to samples

##Import Data & Clean

0.1 Assessing standard Curves

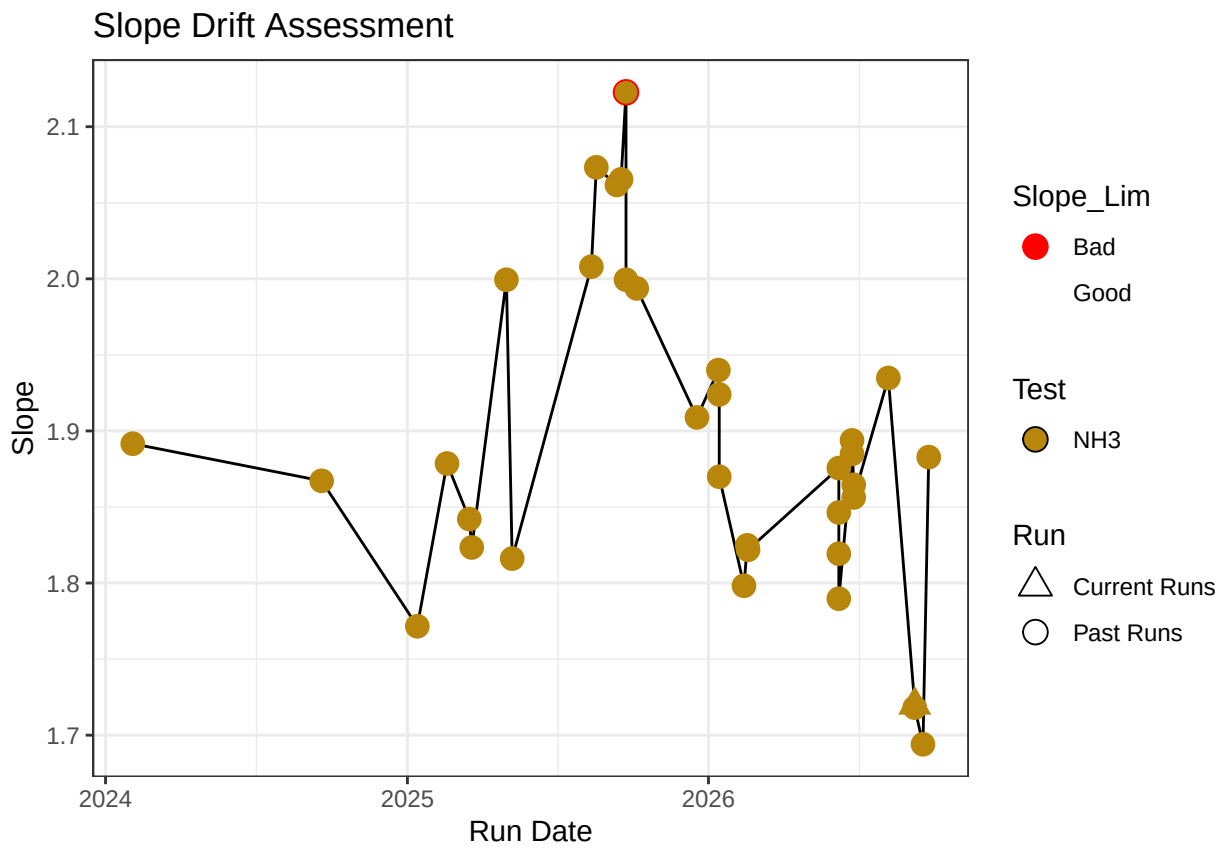
```
## `geom_smooth()` using formula = 'y ~ x'
```

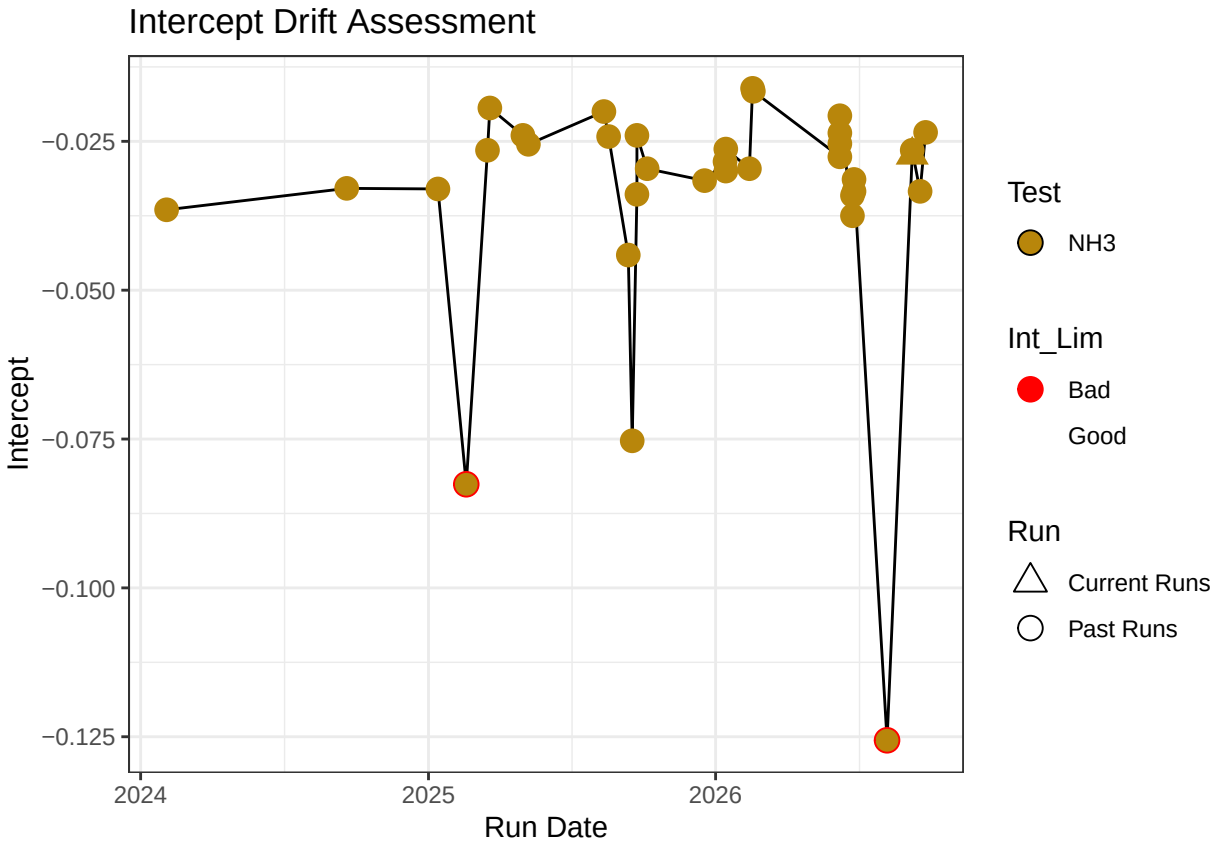


Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	09/08/2026	R2_Pass	0.9998318	1.719787

0.2 Update Standard Log

```
## `summarise()` has grouped output by 'Test', 'Intercept', 'Slope', 'R_Squared',
## 'User'. You can override using the `.groups` argument.
```


Slope Drift Assessment
Slope
2.1
2.0
1.9
1.8
1.7
2024
2025
2026
Run Date
Slope_Lim
Bad
Good
Test
NH3
Run
Current Runs
Past Runs



0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: P12-40"
```

```
## [1] "Diluted Samples: P12-40"
```

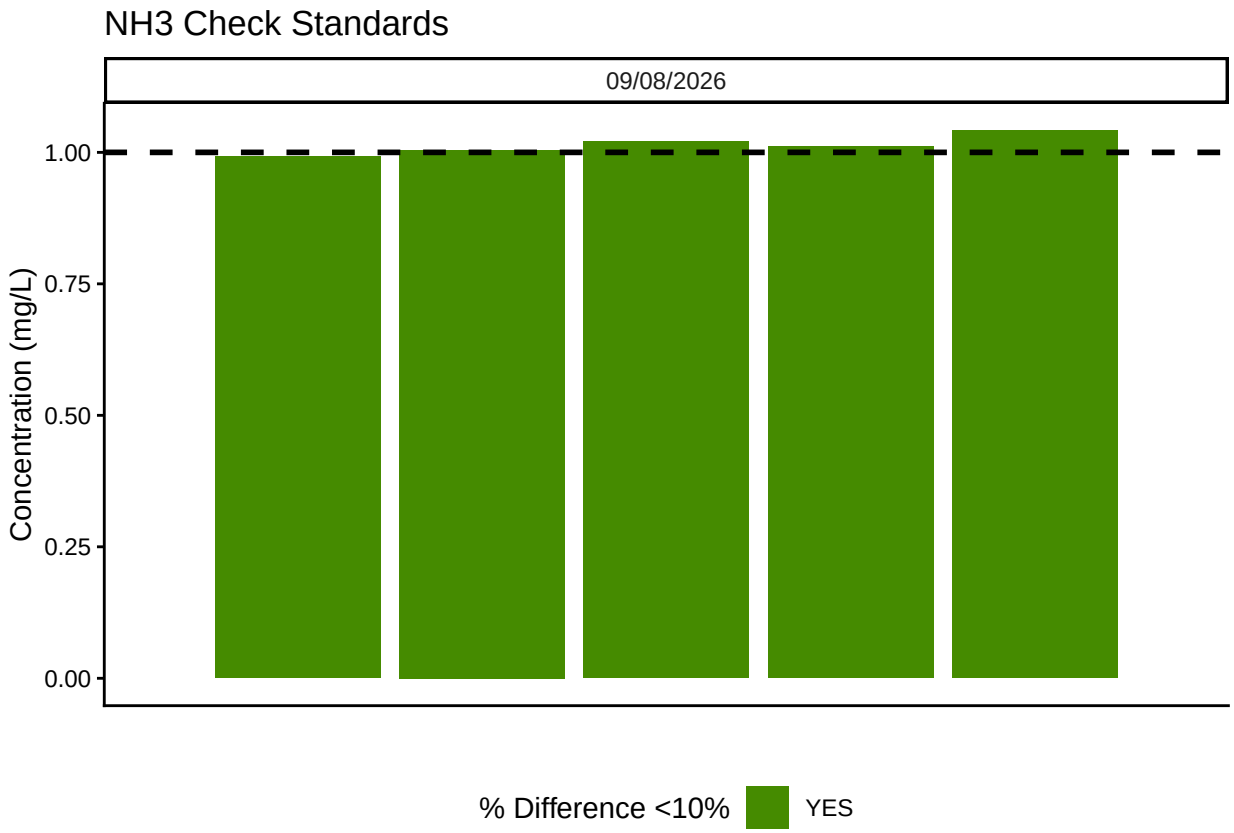
```
## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	09/08/2026	Performance Check Within 25% - PROCEED	1.138257	1.034

0.5 Analyze the Check Standards

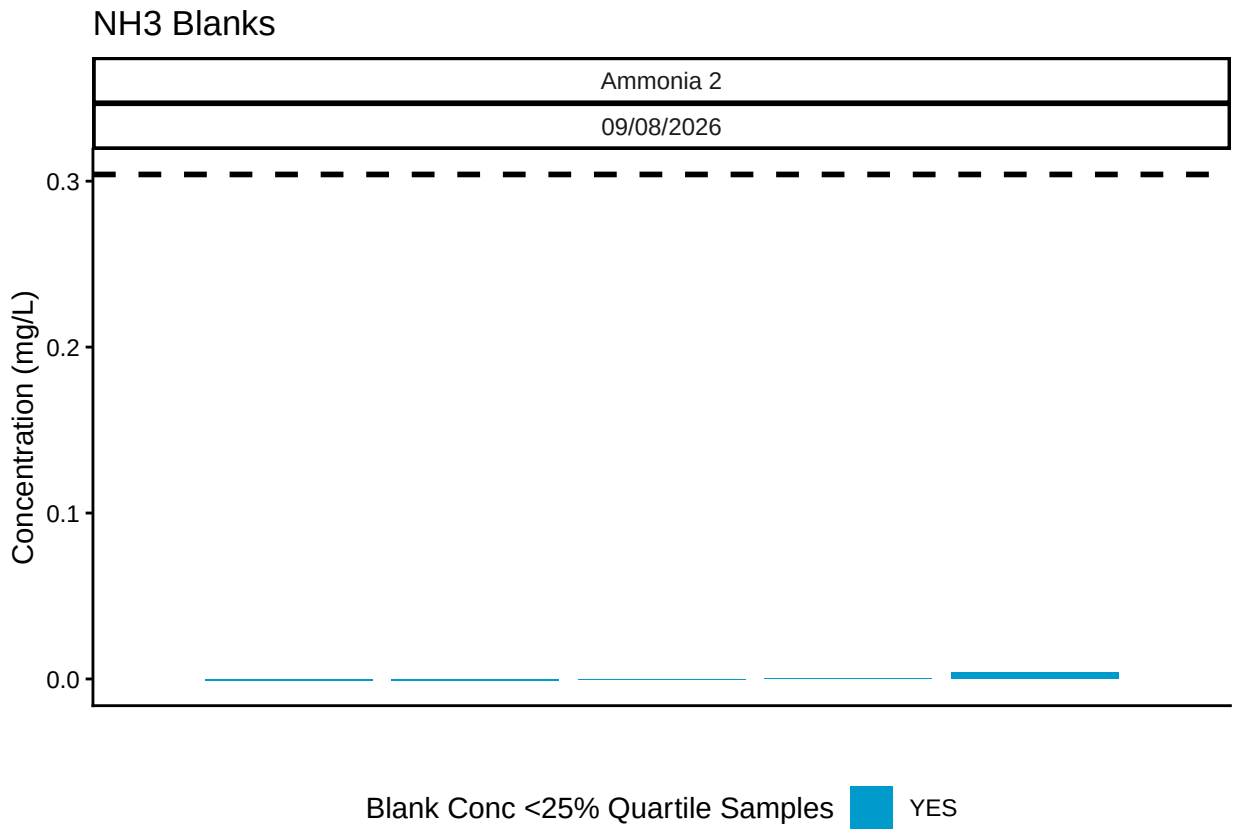
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	09/08/2026	RSV WITHIN RANGE - PROCEED	0.0183453	0.25



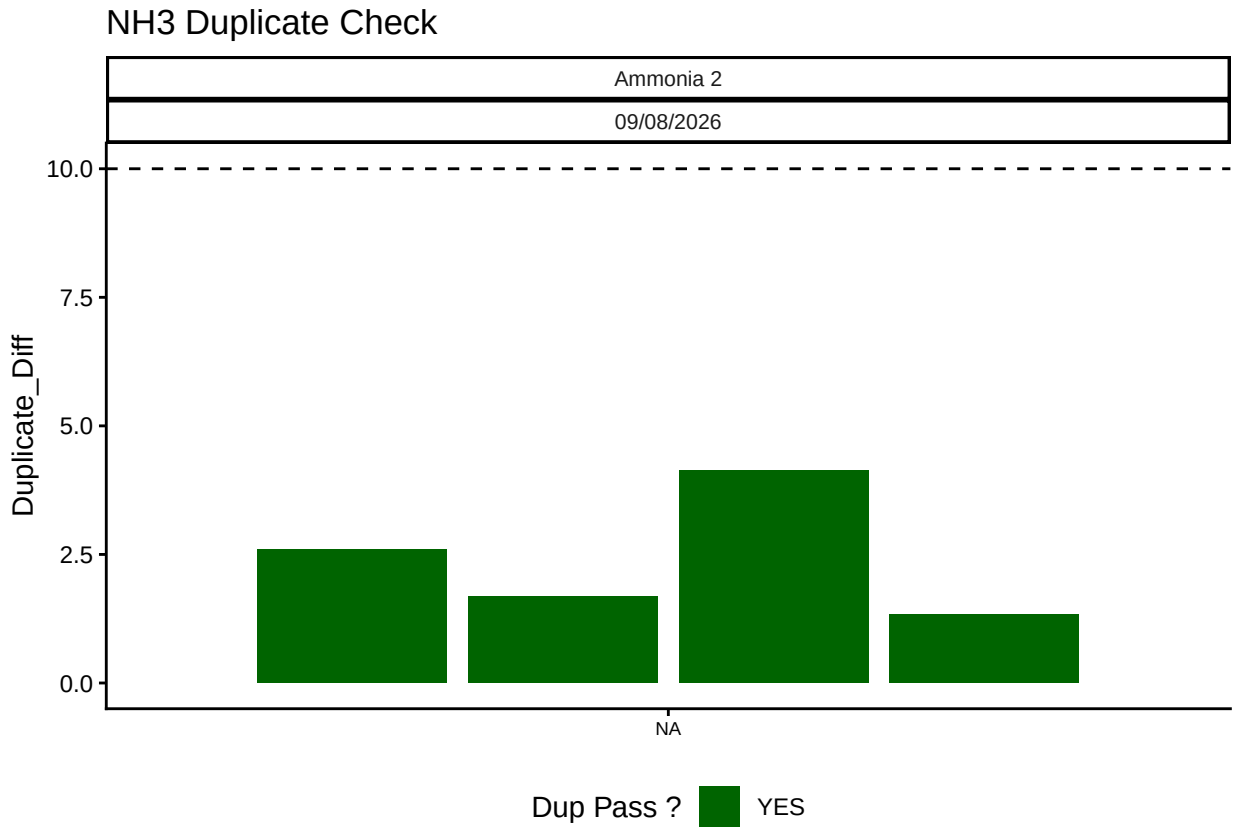
Test	Run_Date	CHK_Flag
Ammonia 2	09/08/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blk_Conc	Quantile_25
Ammonia 2	09/08/2026	>60% of Blanks Pass - PROCEED	0.0005128	0.3040558

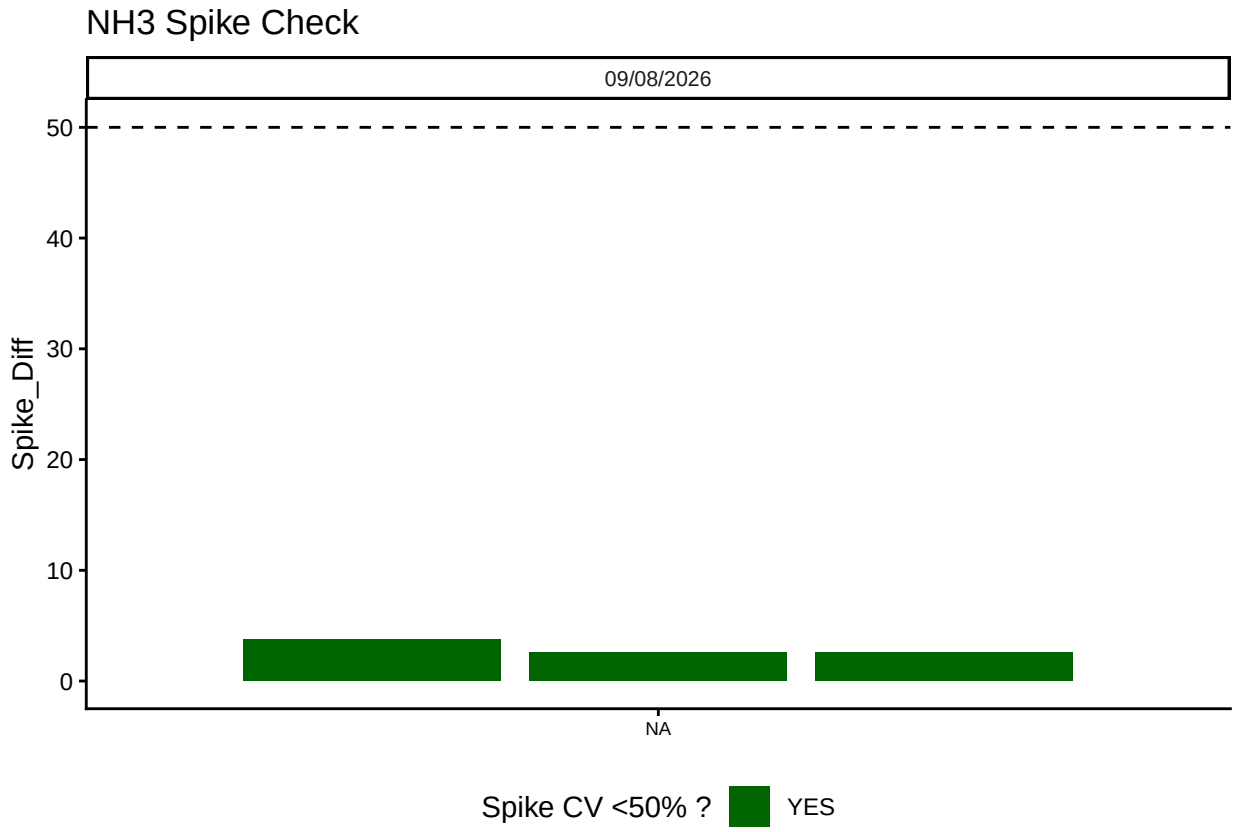


0.7 Analyze Duplicates



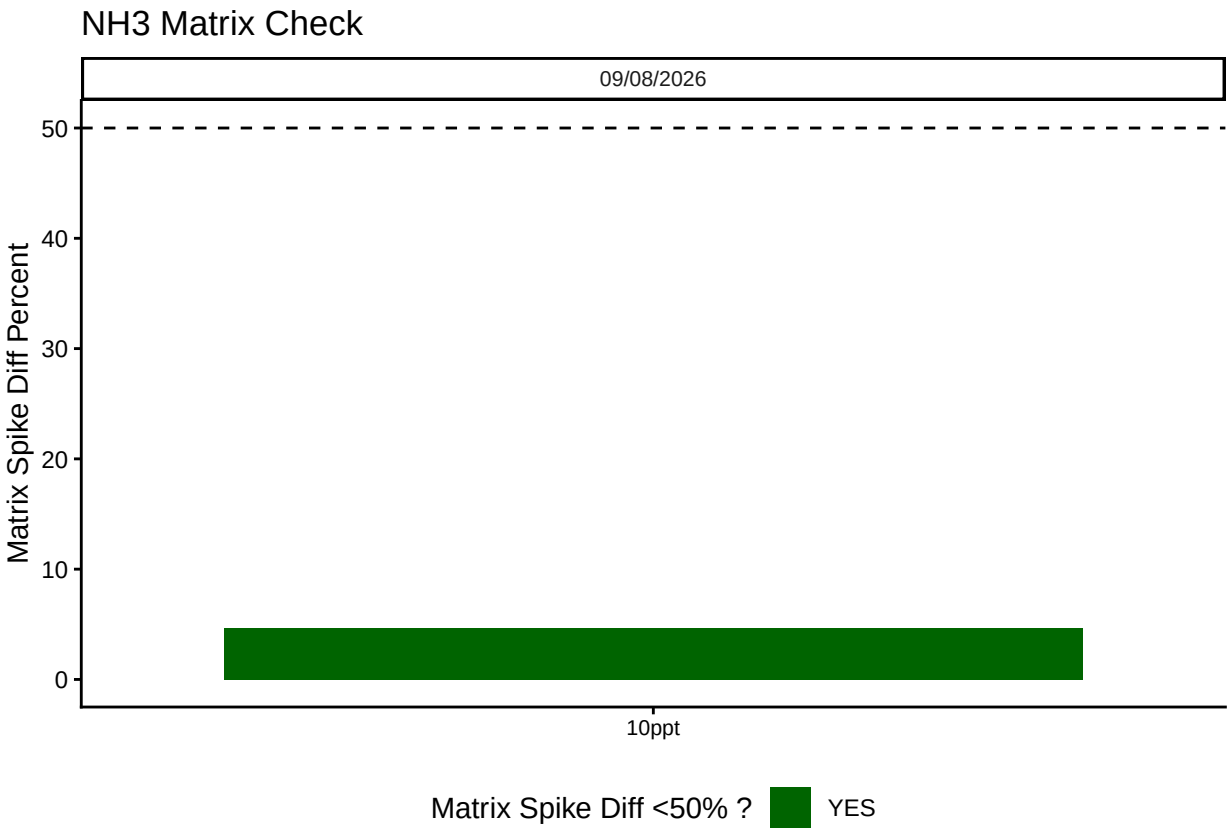
Test	Run_Date	Dup_Flags
Ammonia 2	09/08/2026	>60% of Dups Pass - PROCEED

0.8 Spikes



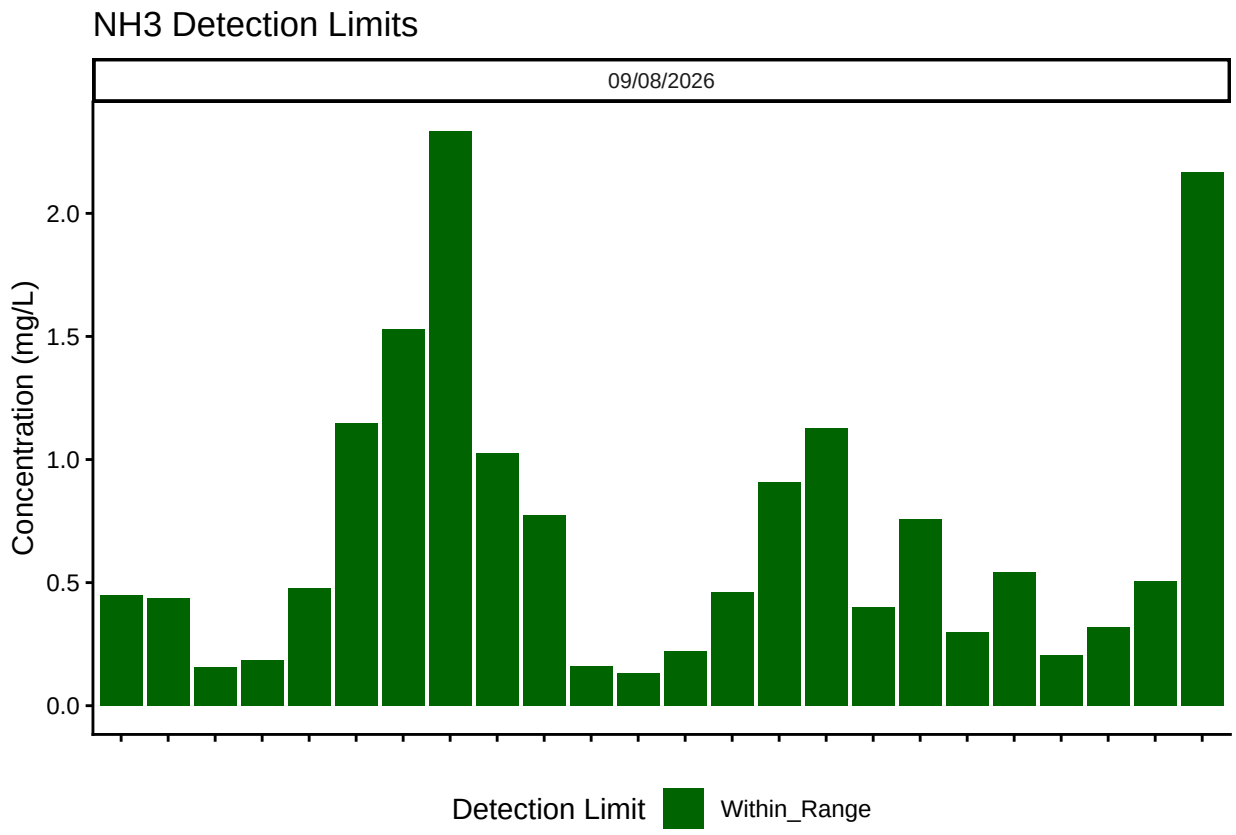
Test	Run_Date	Spike_Flags
Ammonia 2	09/08/2026	>60% of Spikes have a Diff <50% - PROCEED

0.9 Matrix Effects



Test	Run_Date	Matrix_Flags
Ammonia 2	09/08/2026	Matrix Has Diff <50% - PROCEED

##Sample Flagging - Within range of standard curve

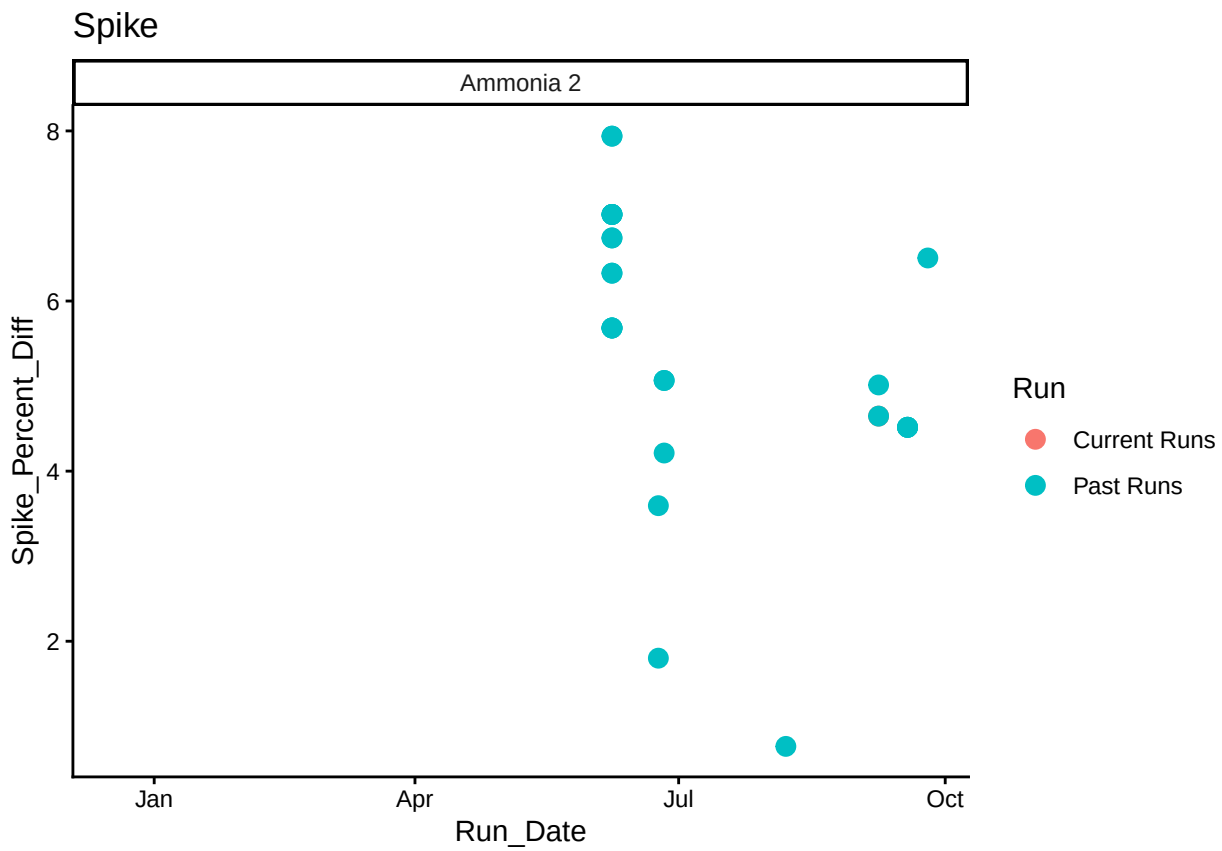


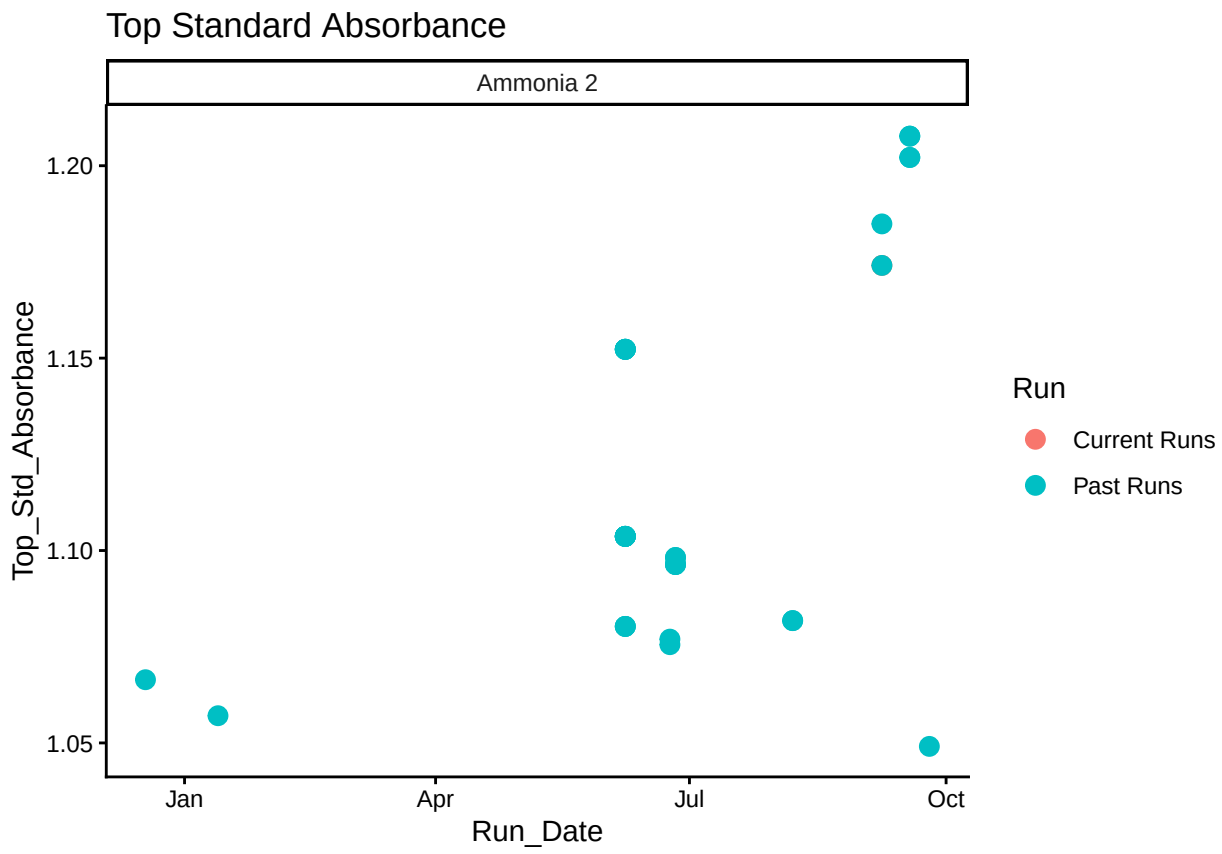
##Add QAQC Flags

0.10 Create Log for QAQC

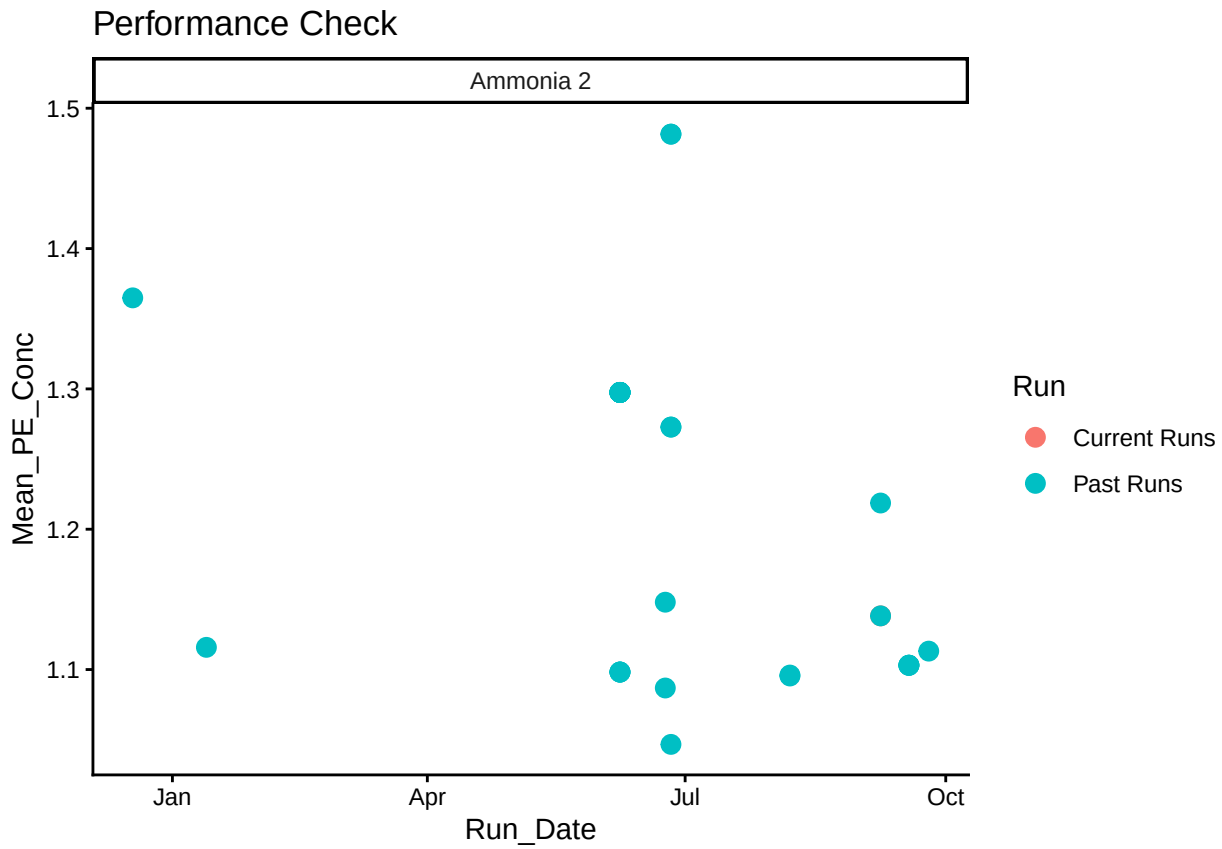
```
## `summarise()` has grouped output by 'Test', 'Run_Date', 'Top_Std_Absorbance',
## 'Mean_PE_Conc'. You can override using the `.groups` argument.
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
```


Spike
Ammonia 2
Spike_Percent_Diff
8
6
4
2
Jan
Apr
Jul
Oct
Run_Date
Run
Current Runs
Past Runs




```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



##Write QAQC Log

0.11 Merge samples with Metadata & check all samples are present

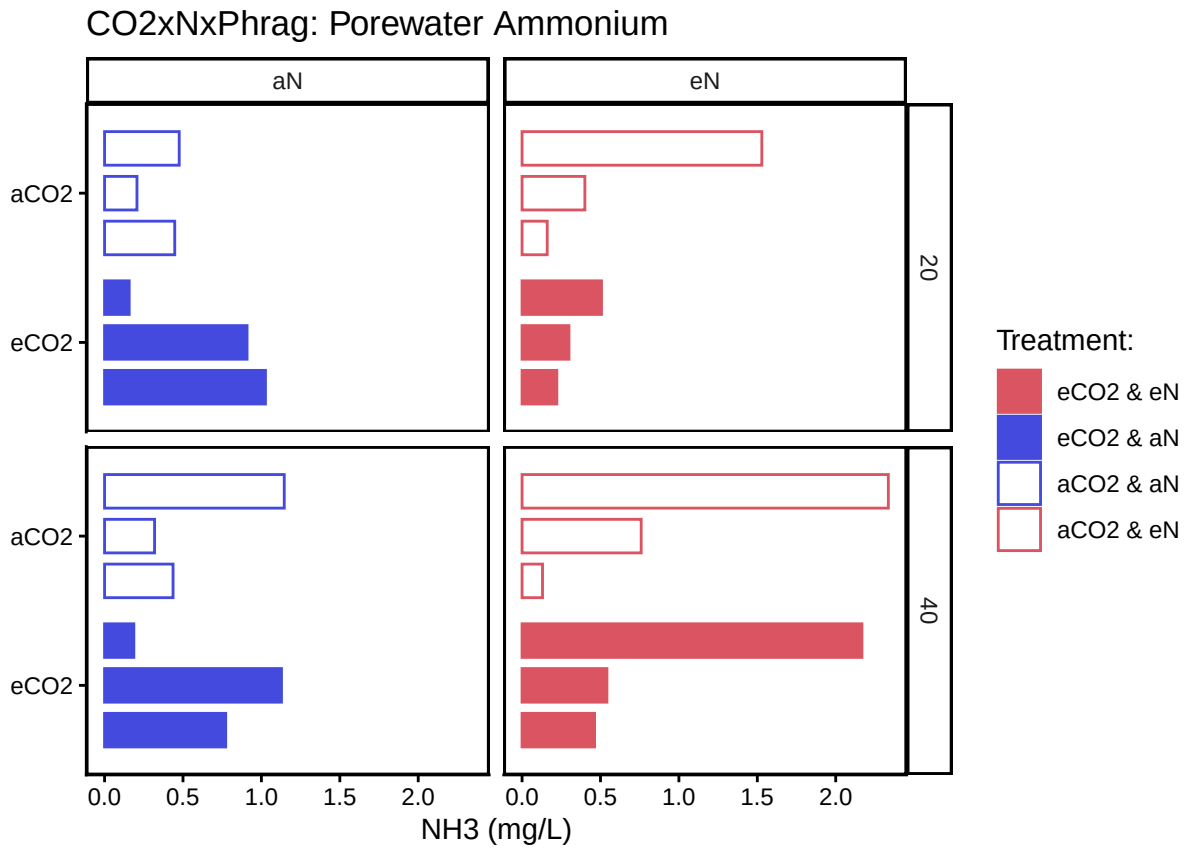
```
## Merge Samples with Metadata

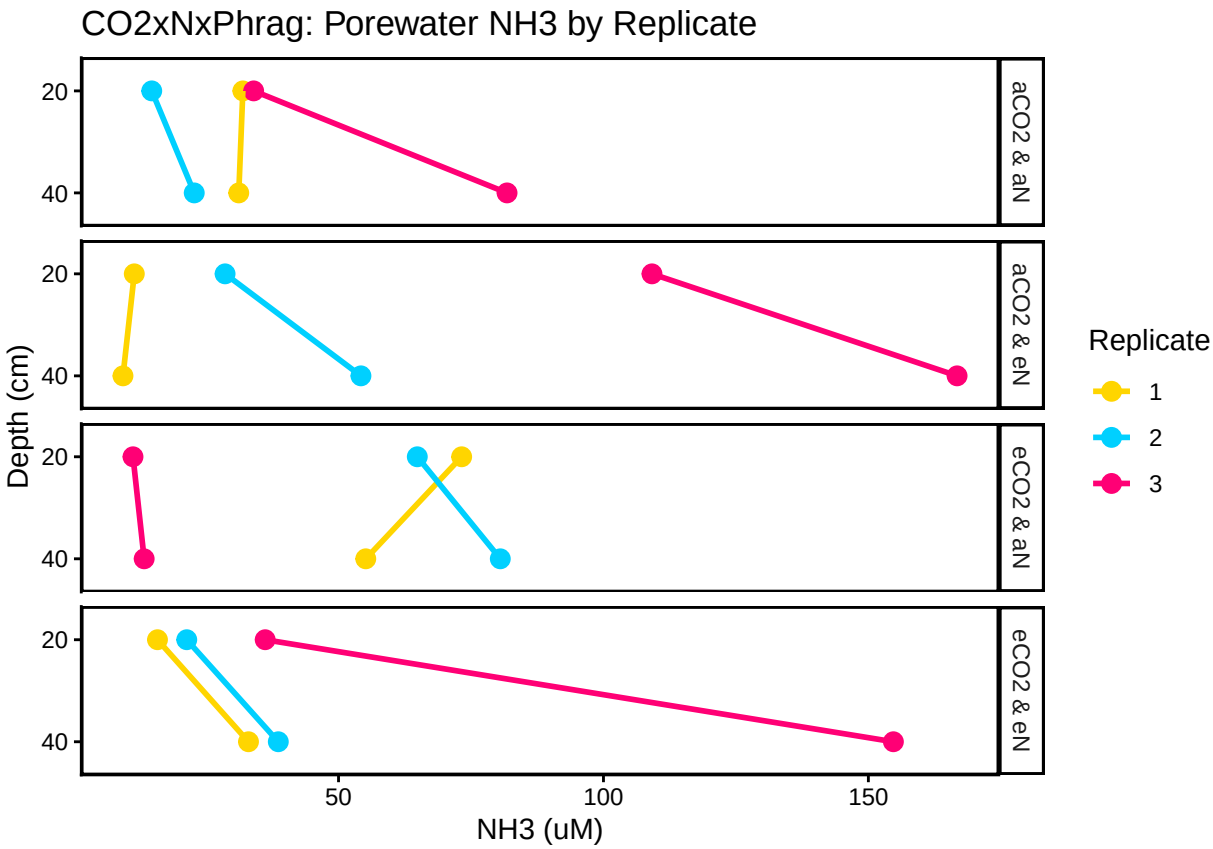
## All sample IDs are present in data
```

0.12 Organize Data

0.13 Visualize Data

```
## Visualize Data
```





##Export Data

#end